Deep Learning for Automatic Cancer Segmentation and Classification in 3D CT Scans

Keishi Suzuki[0009−0006−0217−6511]

University of Toronto, Toronto ON, Canada
keishi.suzuki@mail.utoronto.ca

Abstract. Accurate segmentation and classification of pancreatic cancer lesions in 3D Computed Tomography (CT) scans are essential for early diagnosis and personalized treatment. In this work, a multi-task deep learning framework built upon the nnU-Net v2 architecture was leveraged to simultaneously perform organ segmentation and lesion subtype classification. A shared 3D ResNet-M Encoder was used to extract hierarchical spatial features, feeding into two specialized heads: the 3D ResNet Decoder for the segmentation task and a cross attention pooling head for the classification task. Methods such as weighted cross entropy and down-weighted classification loss term were used to address class imbalance and overfitting issues for classification tasks. This method achieved an average score Dice Similarity Coefficient (DSC) of 0.93 for the whole pancreas and a lesion DSC of 0.66 on the validation set of 36 heterogenous cases. For the classification task, the model achieved a macro-average F1-score of 0.62 and a balanced accuracy of 0.62.

Keywords: Cancer segmentation · Multi-task learning · Pancreatic lesion.

1 Introduction

Approximately 8.4% of all cancer deaths in the US are pancreatic cancer cases and the 5 year relative survival rate is 13.3% [5]. To tackle this, high quality lesion segmentation and subtype classification are essential for early detection and personalized treatment of pancreatic cancer. However, technical challenges such as multi-task difficulty, lesion segmentation difficulty, and data scarcity make this difficult to carry out. Using a shared encoder creates a bias towards segmentation due to more effective pixel level signals, compared to the one signal per volume for the classification head, leading to under-training without specific adjustments.

nnU-Net [2] has established itself as a strong out of the box tool for 3D image segmentation, automatically adapting preprocessing and architecture to fit the specific dataset. Recent work has extended the nnU-Net framework for multi-task settings [6,1] by incorporating cross attention pooling for joint segmentation and classification.

Building on these advances, we aimed to design a single multi-task model that

performs both pancreas/lesion segmentation and lesion subtype classification, while explicitly addressing the imbalance in supervisory signal and the overfitting risk of the classification head. We designed and implemented a multi-task framework based on the nnU-Net v2 [2], with a shared 3D encoder, a segmentation decoder, and a cross-attention pooling head [6] for lesion subtype classification. We addressed class imbalance and overfitting for the classification task through methods such as weighted cross-entropy, down-weighting the classification loss, and dropout. We reported organ DSC, lesion DSC, and classification metrics on a held-out validation set (36 cases), demonstrating that joint training can achieve reasonable segmentation and subtype performance in one model.

2 Method

Our method comprises three stages: preprocessing of 3D CT volumes, a multi-task network that performs pancreas/lesion segmentation and lesion subtype classification, and post-processing of the segmentation output. Figure 1 shows the overall pipeline.

2.1 Preprocessing

Preprocessing follows the nnU-Net pipeline [2]. Following the nnU-Net documentation, resampling and target spacing are determined by the dataset fingerprint. For intensity normalization, we use the standard nnU-Net CT scheme: foreground voxel intensities (all classes except background and ignore) are collected across the training set; the mean, standard deviation, and 0.5/99.5 percentiles are computed; each volume is clipped to the percentiles and then normalized by subtraction of the mean and division by the standard deviation.

2.2 Proposed Method

Our framework is built upon the nnU-Net v2 architecture [2], using a 3D U-Net as a shared encoder for both segmentation and classification (Figure 1). The encoder consists of six stages of 3D convolutional blocks with instance normalization and LeakyReLU, where each stage reduces the spatial resolution according to the strides defined in the nnU-Net plans and increases the number of feature channels (32, 64, 128, 256, 320, 320). The decoder follows the standard nnU-Net design: at each level, a transpose-convolution upsamples the feature maps, which are concatenated with the corresponding encoder features via skip connections, followed by two convolutional blocks. A $1 \times 1 \times 1$ convolutional layer maps the decoder output to voxel-wise logits for pancreas and lesion segmentation.

To support lesion subtype classification, we wrap this base network in a multi-task architecture. For classification, we branch off from the deepest encoder feature map (bottleneck) and apply a cross-attention pooling head followed by a lightweight classifier. Concretely, the bottleneck feature map of shape (B, C, D', H', W') is reshaped to a sequence of spatial tokens (B, S, C), where

$S = D'H'W'$. A small set of learnable query tokens attends to these spatial tokens via multi-head attention, producing a fixed-size representation that is flattened and passed through a LayerNorm and a linear layer to output logits for the three lesion subtypes. This design allows the classifier to focus on discriminative regions while keeping the head compact, which is beneficial given the limited number of labeled cases.

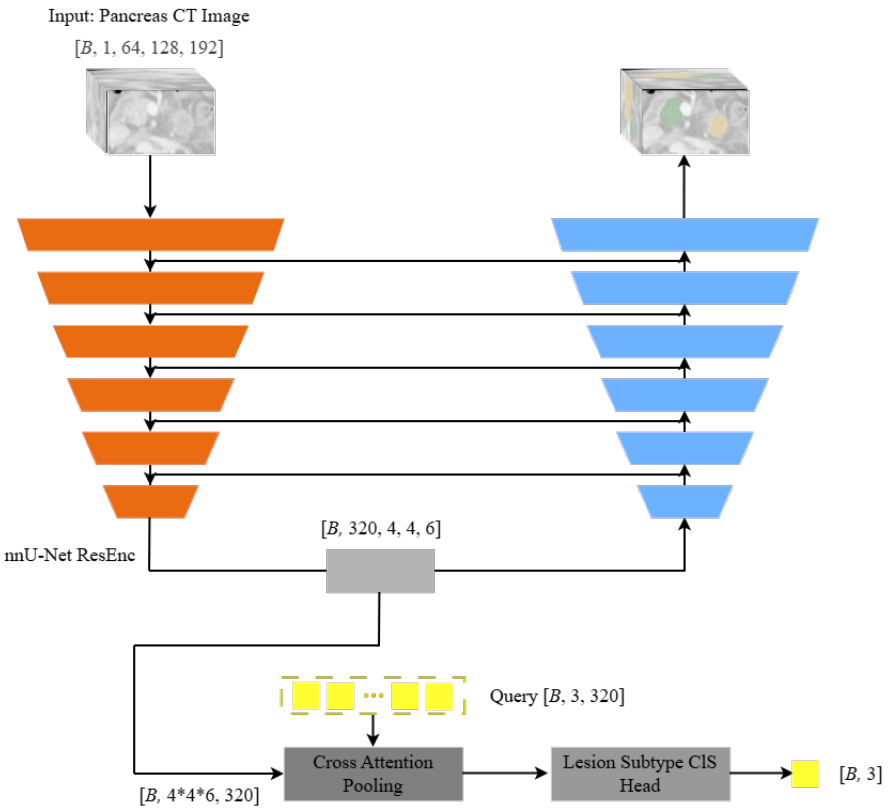

Fig. 1. Network architecture of our proposed method. A 3D U-Net encoder is shared between two heads: the standard nnU-Net decoder for pancreas and lesion segmentation, and a cross-attention pooling–based head for lesion subtype classification.

Loss functions. We jointly optimize the segmentation and classification tasks. For segmentation, we use the standard nnU-Net compound loss, i.e. the sum of soft Dice loss and voxel-wise cross-entropy loss, which has been shown to be robust across a wide range of medical image segmentation tasks [3]. For the classification head, we employ a class-weighted cross-entropy loss to address the imbalance between the three subtypes (62, 106, and 84 training cases, respec-

tively). Class weights are computed as inverse class frequencies, normalized to sum to the number of classes. The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}}^{\text{Dice+CE}} \; + \; \lambda_{\text{cls}} \, \mathcal{L}_{\text{cls}}^{\text{wCE}},$$

where $\lambda_{\text{cls}} = 0.1$ scales the contribution of the classification loss so that segmentation remains the primary task while the classifier is regularized by the shared encoder. We also apply dropout in the classification head and gradient clipping to further reduce overfitting and stabilize optimization.

2.3 Post-processing

During training and validation, we did not apply any task-specific post-processing beyond the default nnU-Net pipeline. For segmentation, the 3D U-Net decoder outputs voxel-wise logits, which are converted to discrete labels by applying softmax followed by an argmax over the class dimension. No additional operations were used. For the classification task, the lesion subtype prediction is obtained by taking the argmax over the three-class logits produced by the cross-attention pooling head. Thus, the final outputs correspond directly to the network predictions without further post-processing.

3 Experiments

3.1 Dataset and evaluation measures

Dataset The dataset was provided for the purpose of this coding test. It consists of de-identified 3D pancreatic CT scans cropped to regions of interest (ROIs) around the pancreas. Each scan is annotated with two labels: label 1 for normal pancreas and label 2 for pancreatic lesion (label 0 is background). In addition, each case carries a lesion subtype label (subtype 0, 1, or 2) for the classification task. The training split contains 252 cases (62 subtype 0, 106 subtype 1, and 84 subtype 2), the validation split contains 36 cases (9 subtype 0, 15 subtype 1, and 12 subtype 2), and a held-out test split contains images only (no labels). No additional public datasets or pre-trained weights were used.

Evaluation measures Evaluation follows the Metrics Reloaded framework [4], which provides principled recommendations for image analysis validation. All metrics are computed using the MetricsReloaded library.

Segmentation. For both the whole pancreas (any non-background voxel; label > 0) and the lesion (label $= 2$), we compute the following metrics per case and then average over all validation cases:

- **Dice Similarity Coefficient (**DSC**)**: measures volumetric overlap between predicted and reference segmentation.

- **Normalized Surface Dice** (NSD, $\tau = 1\,\text{mm}$): a surface-based metric that is tolerant to small boundary deviations up to 1 mm.
- F_β **score** ($\beta = 2$): a recall-weighted overlap measure that penalizes false negatives more heavily than false positives, appropriate for cases where missing lesion tissue is more critical than over-segmentation.

Classification. For lesion subtype classification, we report the following metrics over all 36 validation cases:

- **Macro-average F1**: the mean per-class F_1 score across the three subtypes, giving equal weight to each class regardless of support.
- **Balanced Accuracy** (BA): the mean per-class recall, which is robust to class imbalance.
- **Positive Likelihood Ratio** (LR+) **per class**: measures how much more likely a positive prediction is in true positive cases versus false positive cases.
- **Expected Calibration Error** (ECE^{KDE}), **Class-Wise ECE** (CWCE), and **Reliability Brier Score** (RBS): calibration metrics that assess how well the predicted class probabilities reflect true outcome likelihoods.

3.2 Implementation details

Environment settings The development environments and requirements are presented in Table 1.

Table 1. Development environments and requirements.

System	Windows 11
CPU	Intel(R) Core(TM) i7-14650HX (2.20 GHz)
RAM	16×2GB; 5600MT/s
GPU (number and type)	One NVIDIA GeForce RTX 4070 Laptop GPU 8GB
CUDA version	11.8
Programming language	Python 3.12.11
Deep learning framework	torch 2.7.1, torchvision 0.22.1
Specific dependencies	See `requirements.txt` in the repository
Code	https://github.com/Kappapapa123/ML-Quiz-3DMedImg

Training protocols This model was trained on fully labeled data (252 training cases with both segmentation masks and subtype labels). For data augmentation, we followed the nnU-Net v2 default pipeline (spatial transforms such as rotation, scaling, elastic deformation, and mirroring; and intensity transforms such as gamma augmentation, brightness/contrast changes, and Gaussian noise). Patches were sampled randomly with foreground oversampling (approximately 33% of patches centred on foreground voxels), as in the default nnU-Net data

loader. For model selection, the best checkpoint was chosen based on the validation mean foreground Dice over training epochs. Further details of the training protocol are summarized in Table 2.

Figure 2 shows the training and validation curves tracked via Weights & Biases. The training loss decreases steadily, while the validation loss plateaus by approximately epoch 600, with residual fluctuations attributable to the small validation set size (36 cases). Figure 3 shows the validation mean foreground Dice and macro-F1 exhibit a similar pattern. Both metrics plateau in the later epochs with no consistent upwards trend, indicating that further training is unlikely to yield meaningful improvement. Based on these convergence characteristics and hardware constraints, training was terminated at 700 epochs.

Fig. 2. Training total loss (left) and validation total loss (right) curves tracked by Weights & Biases. Color changes reflect resumed training from a checkpoint.

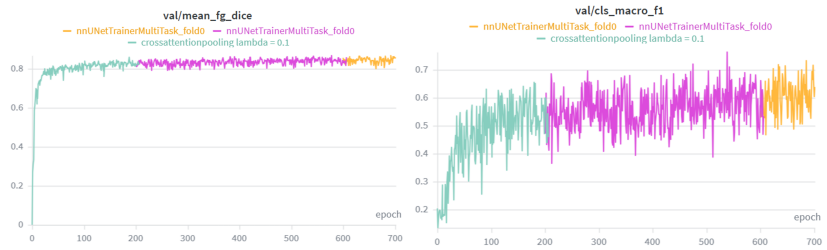

Fig. 3. Validation mean foreground dice (left) and validation macro-F1 (right) curves tracked by Weights & Biases.

Table 2. Training protocols.

Network Initialization	Default (nnU-Net v2 / PyTorch)
Batch size	3
Patch size	$64 \times 128 \times 192$
Total epochs	700
Optimizer	SGD (Nesterov $\mu = 0.99$, weight decay 3×10^{-5})
Initial learning rate (lr)	0.01
Lr decay schedule	Polynomial
Training time	44 hours
Loss function	See Section 2.2

4 Results and discussion

Analyzing the validation results of the pipeline (Table 3), we can observe that the model performs consistently strong on the whole pancreas segmentation, but more unstable on the lesion segmentation.

Table 3. Quantitative evaluation results

Method	Validation Segmentation		Validation Classification	
	Whole Pancreas DSC	Lesion DSC	Macro-F1	BA
Proposed	0.933 ± 0.040	0.662 ± 0.303	0.617	0.617

4.1 Quantitative results on validation set

Tables 4 and 5 summarize the quantitative performance of the proposed method on the validation set ($N = 36$). For segmentation, the model achieves strong whole-pancreas delineation (DSC $= 0.933 \pm 0.040$) while lesion segmentation is considerably weaker (DSC $= 0.662 \pm 0.303$), with the large standard deviation driven by four cases where lesion Dice drops to zero. This difference in performance may be caused by extremely small lesion volumes, causing the model to ignore them due to its small penalty even with class weighted cross entropy. More specific examples are examined in Section 4.2.

For subtype classification, a balanced accuracy and macro-F1 of 0.617 indicate moderate performance; the high $\mathrm{ECE}_{\mathrm{KDE}}$ (0.380) and CWCE (0.239) suggest the classifier is poorly calibrated, likely due to the small training set and class imbalance across the three subtypes.

Table 4. Segmentation performance on the validation set (N=36 cases)

Segmentation area	DSC	NSD ($\tau = 1$ mm)	F_β ($\beta = 2$)
Whole Pancreas	0.933 ± 0.040	0.848 ± 0.105	0.932 ± 0.037
Lesion	0.662 ± 0.303	0.464 ± 0.277	0.654 ± 0.302

Table 5. Classification performance on the validation set (N=36 cases)

Method	Classification metrics					Calibration metrics		
	Macro-F1	BA	LR+ class 0	LR+ class 1	LR+ class 2	ECE^{KDE}	CWCE	RBS
Proposed	0.617	0.617	6.000	4.200	1.750	0.380	0.239	0.823

4.2 Qualitative results on validation set

Figure 4 illustrates qualitative examples of both successful and failed segmentations. In the two strong cases, the model accurately identifies both the whole pancreas and the lesion, consistent with the high Dice scores reported in Section 4.1. In contrast, the failure cases reveal two distinct error types. In one case, the whole-pancreas segmentation remains reasonable, but the lesion is completely missed, yielding a lesion DSC of 0. This may be caused by the small lesion volume. All four validation cases with lesion DSC = 0 have below-median lesion volume, and one case (#227) has an exceptionally small lesion volume (approximately 412 mm^3), suggesting that tiny lesions contribute little to the loss and are prone to being ignored. In another failure case, the model predicts a lesion in an incorrect location and only moderately segments the pancreas, again resulting in a lesion DSC of 0. The cause of this mislocalisation is unclear and may require further case-level investigation.

4.3 Segmentation efficiency results on validation set

Table 6 reports the inference efficiency of the proposed method. All measurements were performed on the hardware described in Table 1 (NVIDIA RTX 4070 Laptop GPU, 8 GB VRAM). Test-time augmentation (TTA) was disabled. On average, segmentation inference takes approximately 16.07 s per 3D CT volume with a peak GPU memory usage of 1006 MB. Classification inference via the cross-attention pooling head adds negligible overhead (< 1 s per volume) since it reuses the encoder features.

Table 6. Segmentation efficiency on the validation set in terms of running time and GPU memory consumption. Evaluation GPU: NVIDIA GeForce RTX 4070 Laptop (8 GB). TTA was disabled.

Case ID	Image Size	Running Time (s)	Max GPU (MB)	Total GPU (MB)
168	(309, 251, 61)	14.90	1017.5	1860.7
184	(181, 109, 50)	14.28	1005.5	1765.7
094	(194, 115, 117)	14.60	994.2	1821.8
154	(195, 125, 132)	21.07	1003.7	2655.3
335	(175, 98, 99)	22.28	994.2	2758.4
088	(171, 132, 54)	15.28	1036.2	2070.4
191	(179, 124, 87)	14.83	1026.8	2109.9
379	(165, 102, 71)	14.57	1003.8	1812.1
Mean ± SD		16.07 ± 2.72	1006.8 ± 10.9	2016.8 ± 350.1

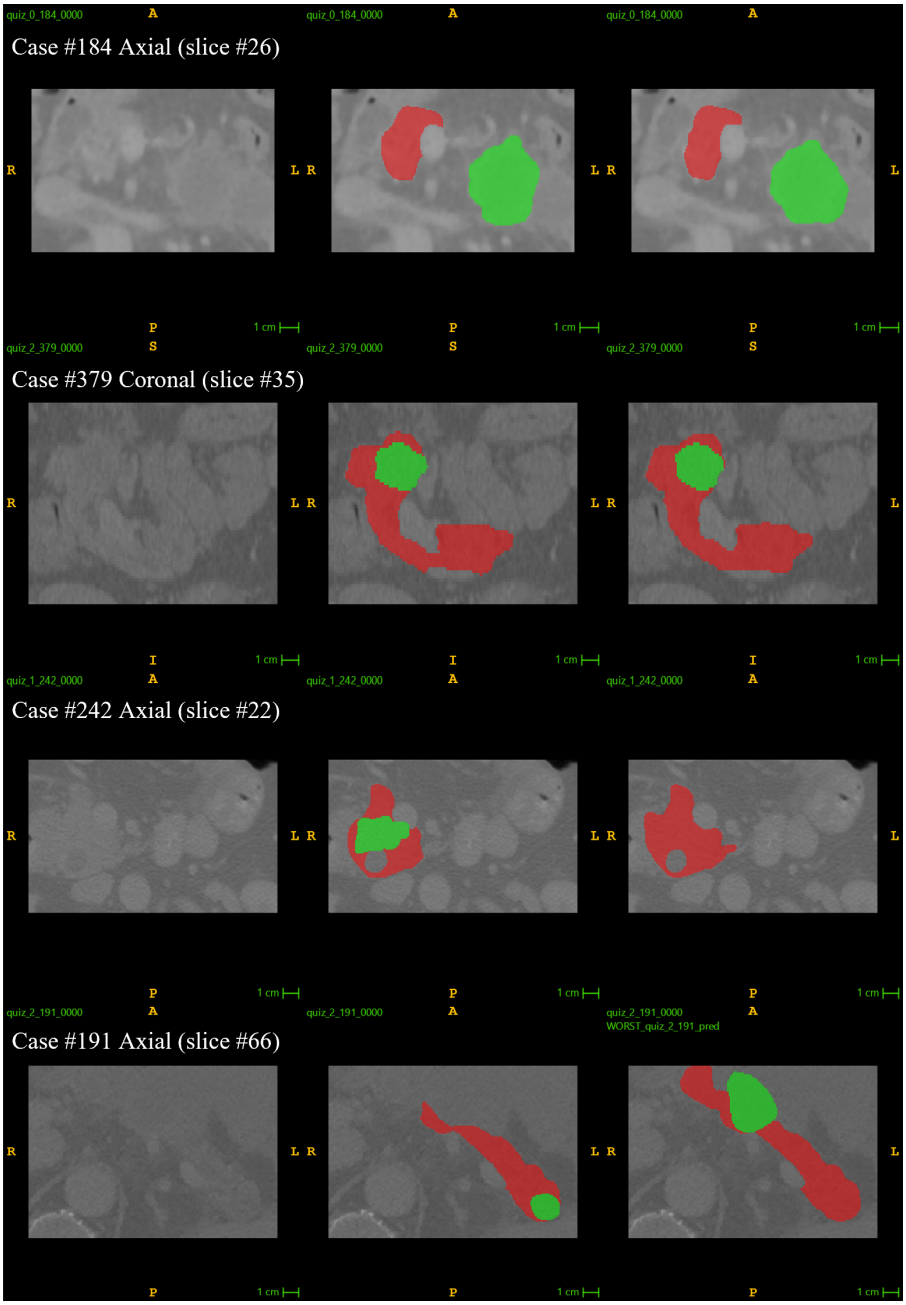

Fig. 4. Qualitative segmentation examples on the validation set. Top: strong performance (cases #184, #379)-whole pancreas (red) and lesion (green) match ground truth. Bottom: failure cases (#242, #191)-#242 shows good pancreas but missed lesion (very small volume); #191 shows incorrect lesion location and moderate pancreas. For each case, left to right: original CT, ground truth mask, predicted mask.

4.4 Limitation and future work

This work has several limitations. First, lesion segmentation performance is substantially weaker and less reliable than whole pancreas segmentation. Although the average lesion DSC score is moderate, we observe large variance and several catastrophic failures with Dice = 0, particularly in cases with very small lesion volumes or completely mislocalised predictions. Second, subtype classification achieves only moderate macro-F1 and balanced accuracy (just above 0.6), and the calibration metrics ($\mathrm{ECE}_{\mathrm{KDE}}$, CWCE) indicate poorly calibrated probabilities. These issues are likely caused by the small and class-imbalanced validation set, but nonetheless limit the clinical usefulness of the current classifier.

Future work will focus on improving lesion segmentation performance and robustness. Possible directions include losses and sampling strategies that place greater emphasis on small lesions (e.g. small-lesion oversampling). For classification, we plan to explore larger and more balanced training cohorts and stronger classification heads.

5 Conclusion

We presented a multi-task deep learning framework for simultaneous pancreatic segmentation and lesion subtype classification, built upon the nnU-Net v2 architecture with a cross-attention pooling classification head branching from the shared encoder bottleneck. On a validation set of 36 cases, the model achieves strong whole-pancreas segmentation (DSC $= 0.933 \pm 0.040$) and moderate lesion segmentation (DSC $= 0.662 \pm 0.303$), with subtype classification balanced accuracy and macro-F1 both at 0.617. Our results demonstrate that joint training within a single network can yield competitive organ-level segmentation while providing lesion subtype predictions at no additional inference cost. However, lesion-level performance remains the primary bottleneck, with catastrophic failures on small or ambiguous lesions, and the classifier suffers from poor calibration under limited training data. Addressing these shortcomings through lesion-aware training strategies and improved calibration techniques will be the focus of future development.

Acknowledgements The authors of this paper declare that the segmentation method they implemented for participation in the coding test has not used any pre-trained models nor additional datasets other than those provided by the organizers. The proposed solution is fully automatic without any manual intervention. We thank all data owners for making the CT scans publicly available.

Disclosure of Interests

The authors declare no competing interests.

References

1. Cao, K., Xia, Y., Yao, J., Han, X., Lambert, L., Zhang, T., Tang, W., Jin, G., Jiang, H., Fang, X., Nogues, I., Li, X., Guo, W., Wang, Y., Fang, W., Qiu, M., Hou, Y., Kovarnik, T., Vocka, M., Lu, Y., Chen, Y., Chen, X., Liu, Z., Zhou, J., Xie, C., Zhang, R., Lu, H., Hager, G.D., Yuille, A.L., Lu, L., Shao, C., Zhang, L., Shi, Y., Zhang, Q., Lu, J.: Large-scale pancreatic cancer detection via non-contrast ct and deep learning. Nature Medicine **29**, 3033–3043 (2023) 1
2. Isensee, F., Jaeger, P.F., Kohl, S.A., Petersen, J., Maier-Hein, K.H.: nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods **18**(2), 203–211 (2021) 1, 2
3. Ma, J., Chen, J., Ng, M., Huang, R., Li, Y., Li, C., Yang, X., Martel, A.L.: Loss odyssey in medical image segmentation. Medical Image Analysis **71**, 102035 (2021) 3
4. Maier-Hein, L., Reinke, A., Godau, P., Tizabi, M.D., Buettner, F., Christodoulou, E., Glocker, B., Isensee, F., Kleesiek, J., Kozubek, M., Reyes, M., Riegler, M.A., Wiesenfarth, M., Kavur, A.E., Sudre, C.H., Baumgartner, M., Eisenmann, M., Heckmann-Nötzel, D., Rädsch, T., Acion, L., Antonelli, M., Arbel, T., Bakas, S., Benis, A., Blaschko, M.B., Cardoso, M.J., Cheplygina, V., Cimini, B.A., Collins, G.S., Farahani, K., Ferrer, L., Galdran, A., van Ginneken, B., Haase, R., Hashimoto, D.A., Hoffman, M.M., Huisman, M., Jannin, P., Kahn, C.E., Kainmueller, D., Kainz, B., Karargyris, A., Karthikesalingam, A., Kofler, F., Kopp-Schneider, A., Kreshuk, A., Kurc, T., Landman, B.A., Litjens, G., Madani, A., Maier-Hein, K., Martel, A.L., Mattson, P., Meijering, E., Menze, B., Moons, K.G.M., Müller, H., Nichyporuk, B., Nickel, F., Petersen, J., Rajpoot, N., Rieke, N., Saez-Rodriguez, J., Sánchez, C.I., Shetty, S., van Smeden, M., Summers, R.M., Taha, A.A., Tiulpin, A., Tsaftaris, S.A., Calster, B.V., Varoquaux, G., Jäger, P.F.: Metrics reloaded: recommendations for image analysis validation. Nature Methods **21**, 195–212 (2024) 4
5. National Cancer Institute: SEER cancer stat facts: Pancreatic cancer. https://seer.cancer.gov/statfacts/html/pancreas.html (nd), surveillance, Epidemiology, and End Results Program, accessed 2025 1
6. Shi, P., Huang, H., Chen, J., YanLu, MakeLoveAndPeace: 2nd place solution: RSNA intracranial aneurysm detection. https://www.kaggle.com/competitions/rsna-intracranial-aneurysm-detection/writeups/2nd-place-solution (2025), kaggle competition write-up 1, 2

Table 7. Checklist Table. Please fill out this checklist table in the answer column.

Requirements	Answer
A meaningful title	Yes
The number of authors ($\leq$6)	1
Author affiliations and ORCID	Yes
Corresponding author email is presented	Yes
Validation scores are presented in the abstract	Yes
Introduction includes at least three parts: background, related work, and motivation	Yes
A pipeline/network figure is provided	1
Pre-processing	2
Post-processing	4
The dataset and evaluation metric section are presented	4
Environment setting table is provided	1
Training protocol table is provided	2
Efficiency evaluation results are provided	6
Visualized segmentation example is provided	4
Limitation and future work are presented	Yes
Reference format is consistent.	Yes